Introduction to Economics

Class 9 — Price controls; international trade and tariffs

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Universitat de Barcelona · Group A6

22 October 2026

Quiz 6

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top.

A question marked + can have more than one correct letter: circle all of them.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

Plan for today

10:00	Quiz 6
10:11	Warm-up
10:15	Catalonia caps rents: what will happen?
10:18	The lost triangle, computed
10:29	In pairs: taxes on menús
10:34	A rent cap: a shortage, now and over the years
10:43	Catalonia: what the data say
10:46	At the board: a cap on allotment rents
10:50	Break · Problem set 1 on the front desk
11:00	A price floor: the minimum wage
11:04	Open borders: who wins, who loses
11:13	In pairs: cava at home and abroad
11:19	A tariff: who gains, what is lost
11:27	European Union (EU) car duties; why limit trade; in pairs
11:39	At the board: a tax on driving lessons
11:44	In one line; before Monday 26 October

Warm-up: which elasticity, what number, what word?

4 % fewer tubs of margarine sold. Two different causes:

1. Its price rose 10 %; nothing else changed

2. Incomes rose 10 %; all prices unchanged

In pairs, one minute; then hands per item.

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   price elasticity of demand: $|-4\,\% \div 10\,\%| = 0.4$**, below 1: inelastic**
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1. Its price rose 10 %; nothing else changed
   price elasticity of demand: $|-4\,\% \div 10\,\%| = 0.4$**, below 1: inelastic**
2. Incomes rose 10 %; all prices unchanged
   income elasticity: $-4\,\% \div 10\,\% = -0.4$**, below 0: an inferior good**

In pairs, one minute; then hands per item.

Warm-up: which curve moves? Rooms for students

Rooms for students in Barcelona

1. A new campus brings 2,000 more students
2. Owners turn rooms into tourist lets
3. Because of 2, rents rise: does demand shift too?

In pairs, one minute; then hands per item.

Warm-up: which curve moves? Rooms for students

Rooms for students in Barcelona

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   demand for rooms shifts right
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2. Owners turn rooms into tourist lets
   supply of rooms shifts left
3. Because of 2, rents rise: does demand shift too?
   no: fewer rooms wanted, along demand

In pairs, one minute; then hands per item.

16 March 2024: Catalonia caps the rent on new contracts

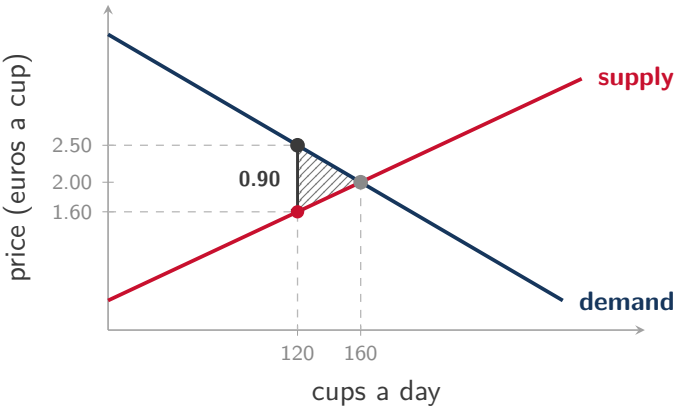


140 towns with a stressed rental market, Barcelona included

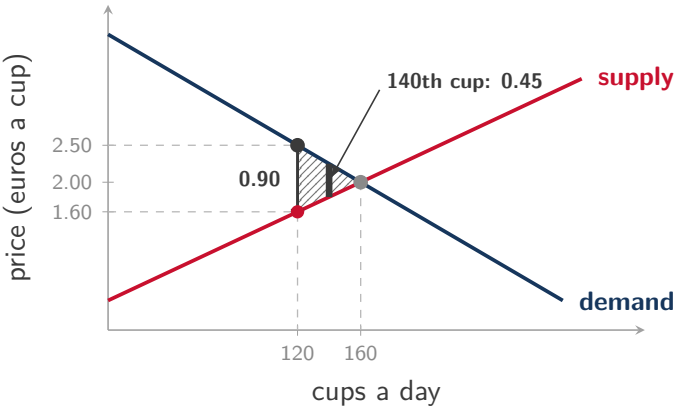
What happens to rents — and to how many flats get rented?

Ley 12/2023 (Spain's housing law); Resolució TER/800/2024
(Diari Oficial de la Generalitat de Catalunya, 14 March 2024), in force 16 March 2024.

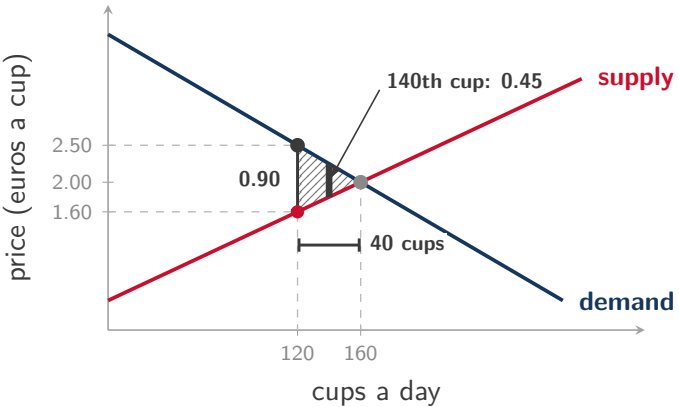
Class 8's lost 18 euros: half the tax times the cups not made



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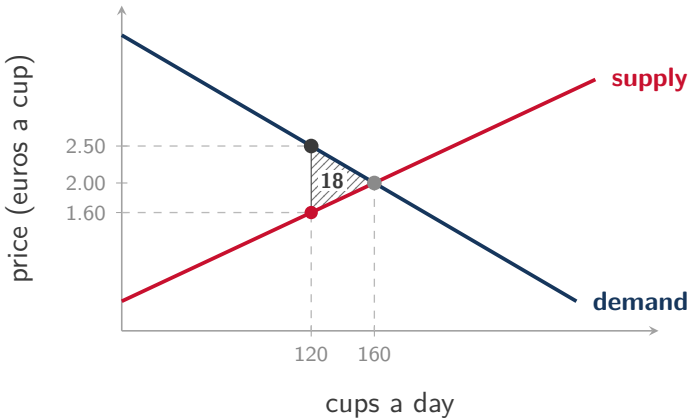


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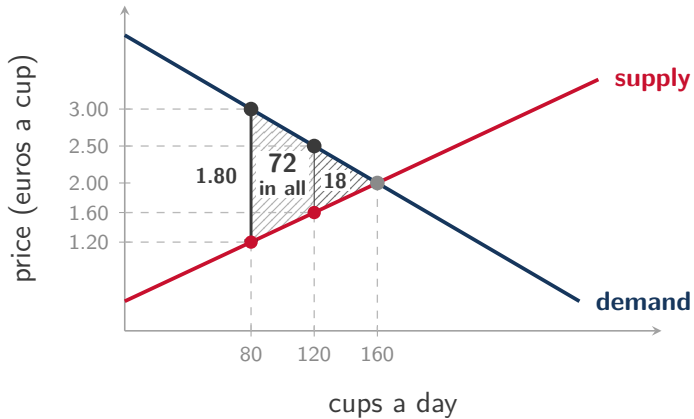


Deadweight loss = ½ × tax × cups lost = ½ × 0.90 × 40 = 18 euros a day.

Twice the tax, four times the loss

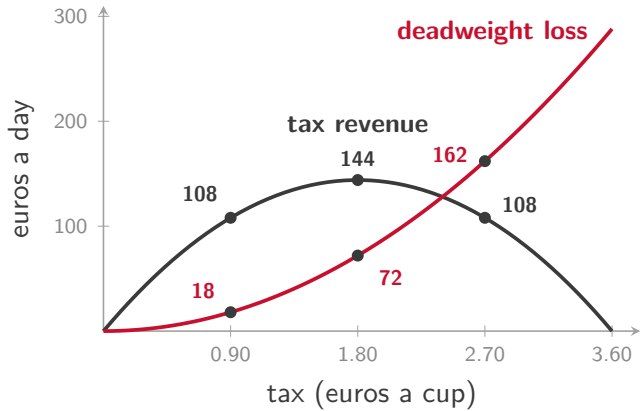


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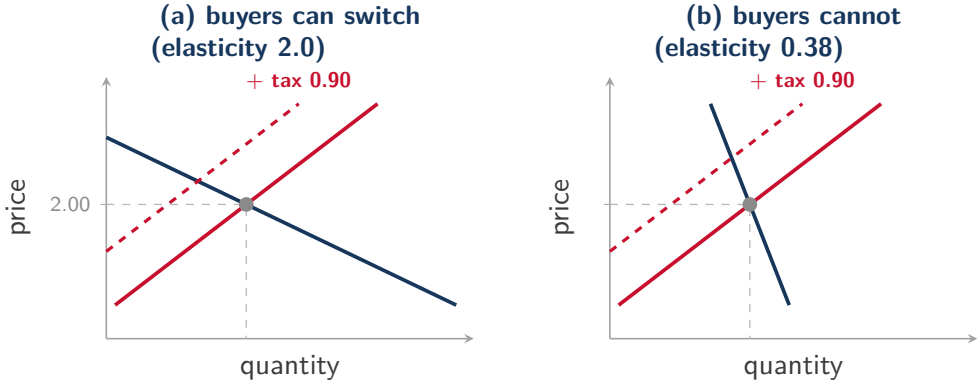
Twice the tax: twice the cups lost, twice the height — four times the loss.

Raise the tax and revenue rises, then falls — the loss keeps growing

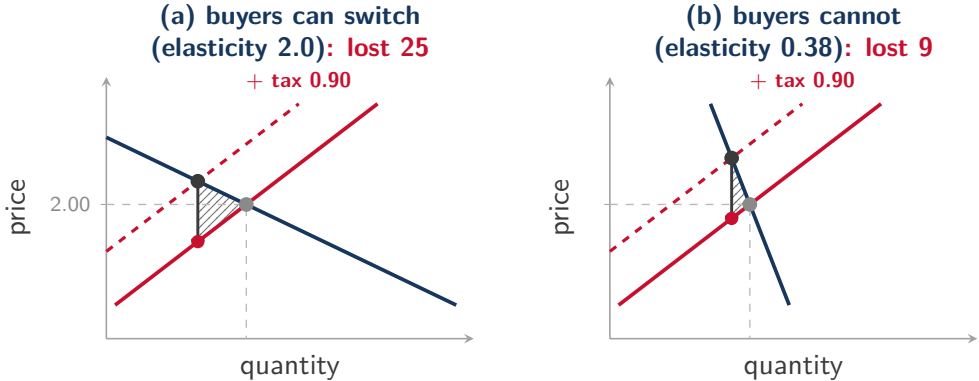


Beyond 1.80 a cup, a higher tax raises less money and loses more.

The same 0.90 tax loses more where buyers can walk away



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The more elastic the market, the more units a tax stops — and the bigger the loss.

In pairs, three minutes: what a tax on menús loses

Same street, same menús

menús a day	the last one is worth	the last one costs
30	15.00	10.00
45	13.50	11.00
60	12.00	12.00

1. Tax 2.50: 45 menús. The lost surplus?
2. Tax 5.00: 30 menús. The lost surplus?
3. Revenue at each tax?

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1. Tax 2.50: 45 menús. The lost surplus?
   ½ × 2.50 × 15 = 18.75 a day
2. Tax 5.00: 30 menús. The lost surplus?
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Same street, same menús

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1. Tax 2.50: 45 menús. The lost surplus?
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2. Tax 5.00: 30 menús. The lost surplus?
   ½ × 5.00 × 30 = 75 a day: four times
3. Revenue at each tax?

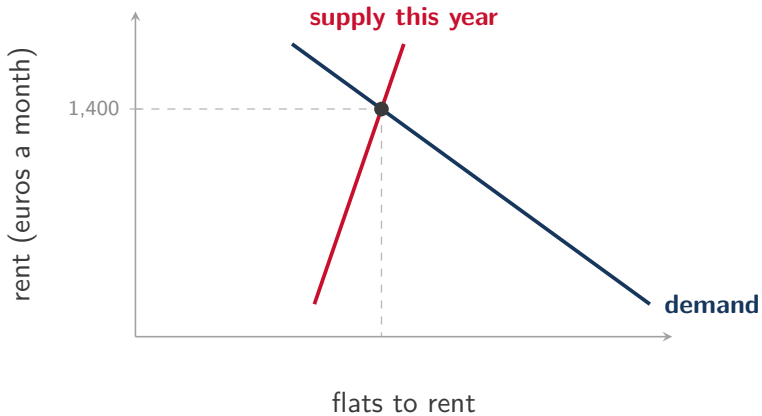
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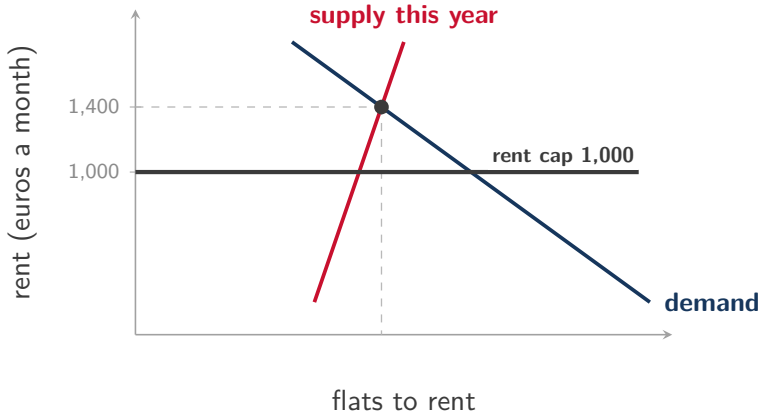
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   ½ × 2.50 × 15 = 18.75 a day
2. Tax 5.00: 30 menús. The lost surplus?
   ½ × 5.00 × 30 = 75 a day: four times
3. Revenue at each tax?
   112.50, then 150: up a third

A rent cap of 1,000 on Gràcia's street: 300 flats wanted, 200 offered



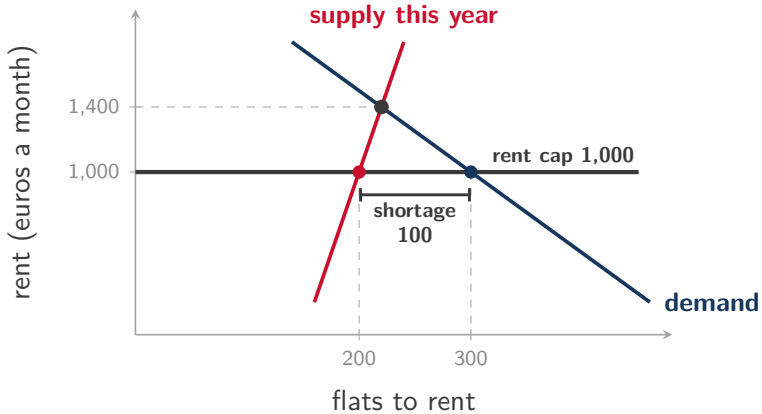
Own numbers (Class 7's Gràcia street); not data.

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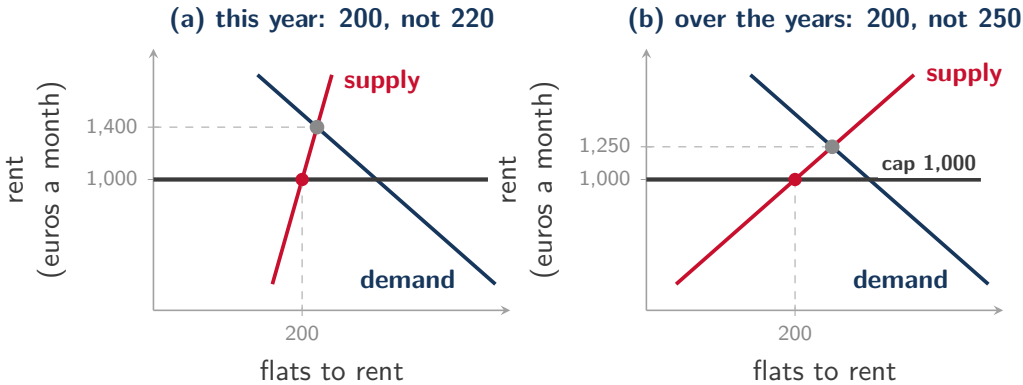
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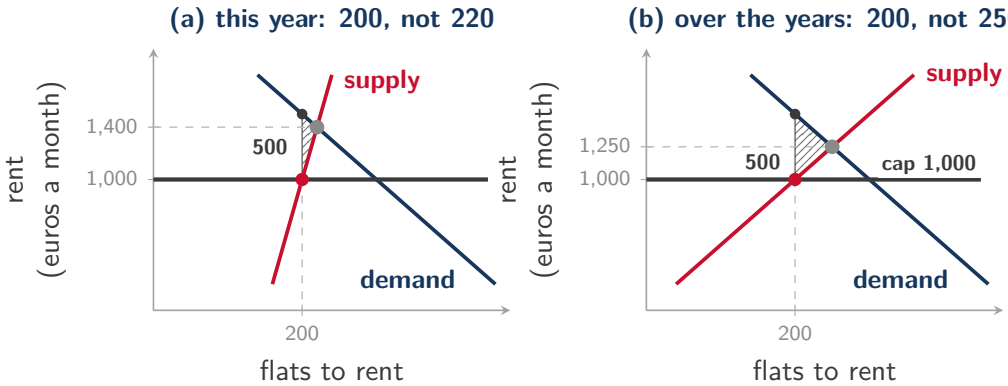
Below the market rent, a cap leaves 100 households searching.

Own numbers (Class 7's Gràcia street); not data.

Over the years the cap costs 50 flats, not 20



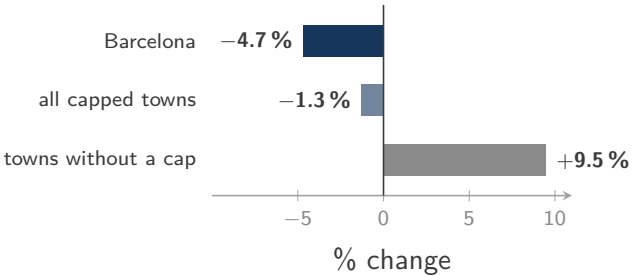
Over the years the cap costs 50 flats, not 20



Lost surplus: ½ × 500 × 20 = 5,000 a month this year;
½ × 500 × 50 = 12,500 over the years.

Catalonia's cap: rents fell — and at first, so did new contracts

Rent on new contracts, early 2024 to early 2026



−17.2 %
new contracts in capped towns, Apr–Jun 2024 vs Apr–Jun 2023

As the model says: lower rents for those who sign, fewer contracts signed.

Generalitat de Catalunya, from Incasòl's register of rental deposits: press notes of 8 October 2024 (April–June 2024) and 13 July 2026 (first quarter of 2026, against the first quarter of 2024).

Practice at the board: a cap on allotment rents

Allotment plots (small vegetable gardens) in a city

Rent P in euros a year · gardeners want $Q = 300 - 2P$ plots
Landowners offer $Q = 4P - 120$ · the council caps the rent at 60

1. No cap: the rent and the number of plots?

2. With the cap: plots rented, plots wanted?

3. Deadweight loss a year?

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   70 euros a year, 160 plots
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   120 rented, 180 wanted: a shortage of 60
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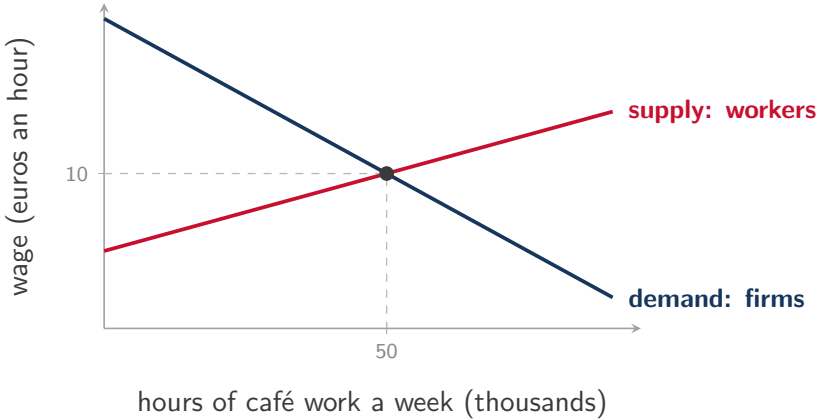
1. No cap: the rent and the number of plots?
   70 euros a year, 160 plots
2. With the cap: plots rented, plots wanted?
   120 rented, 180 wanted: a shortage of 60
3. Deadweight loss a year?
   the 120th plot is worth 90: ½ × (90 − 60) × (160 − 120) = 600 euros

Break

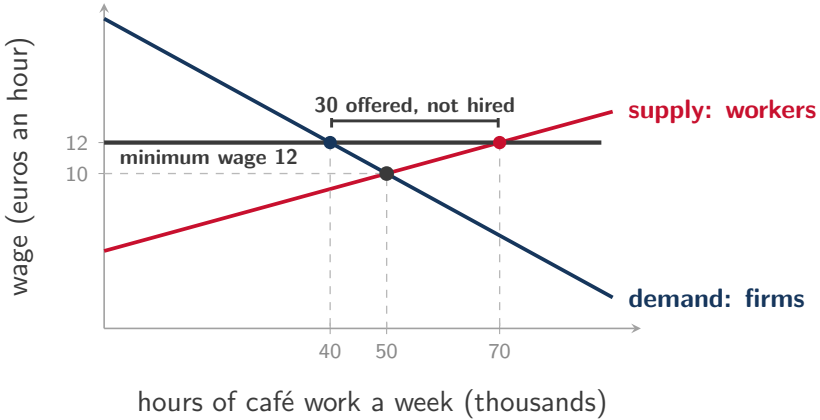
Back at 11:00

Problem set 1: on the front desk as you go out

A minimum wage is a price floor: workers offer 70, firms hire 40

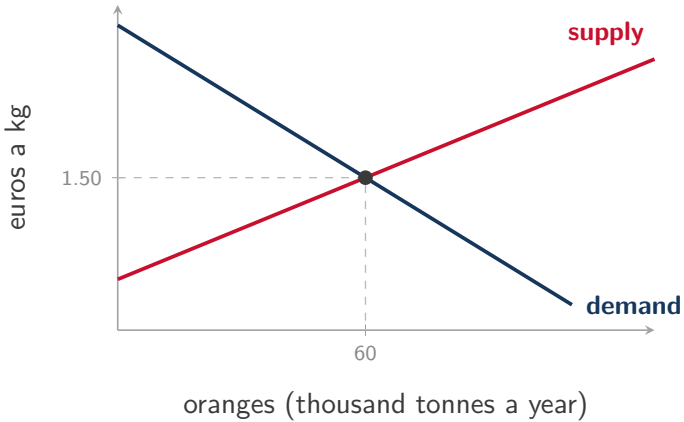


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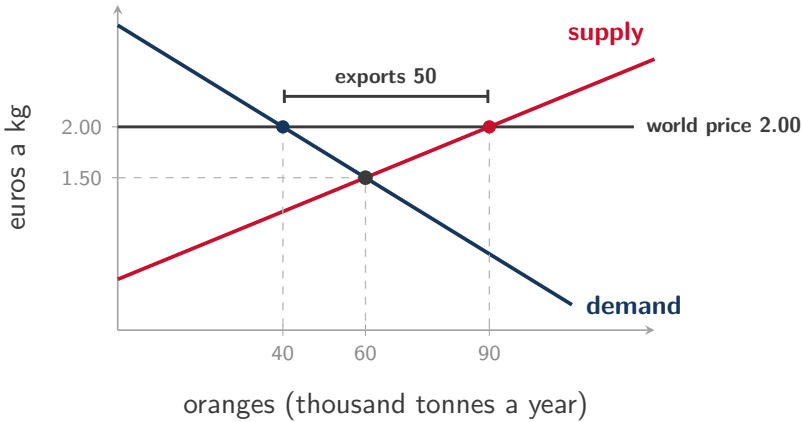


A floor above the market price: sellers offer more than buyers take — here, more hours than firms hire.

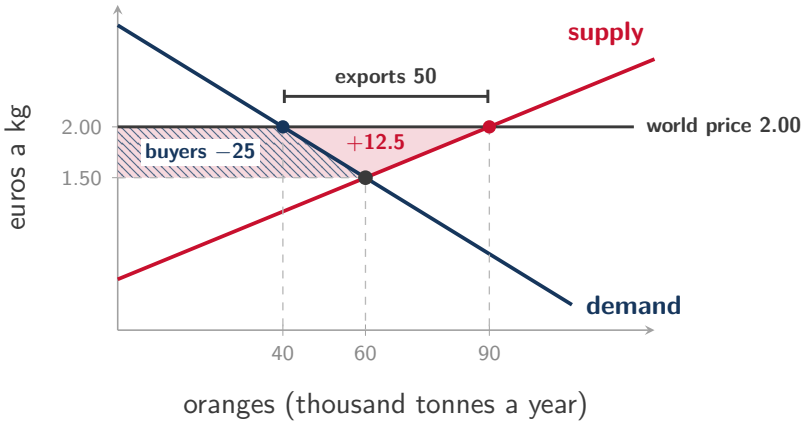
World price above home price: exports; sellers gain, buyers lose less



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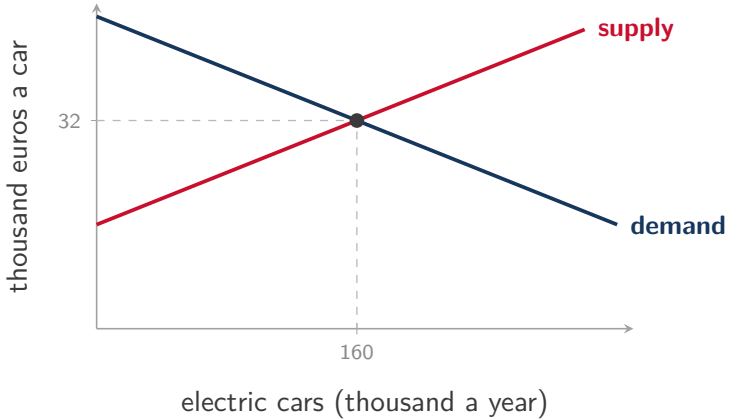
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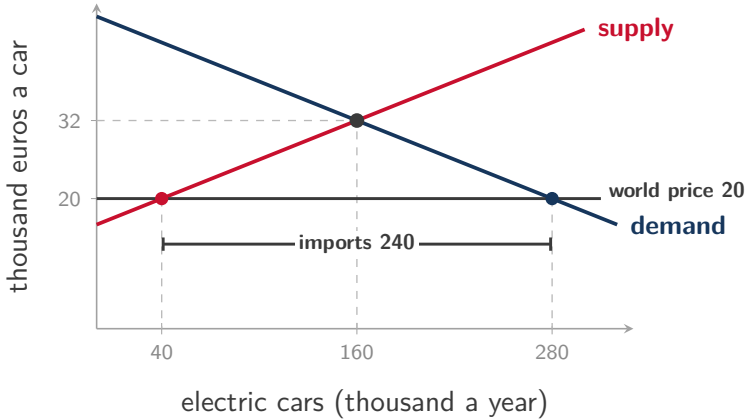
Sellers gain 37.5, buyers lose 25: the country gains 12.5 million euros a year.

1 thousand tonnes = 1 million kg, so euros a kg × thousand tonnes = million euros.

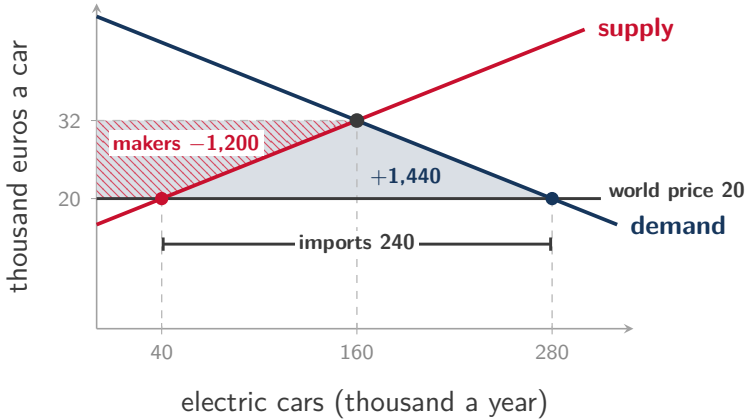
World price below home price: imports; buyers gain, makers lose less



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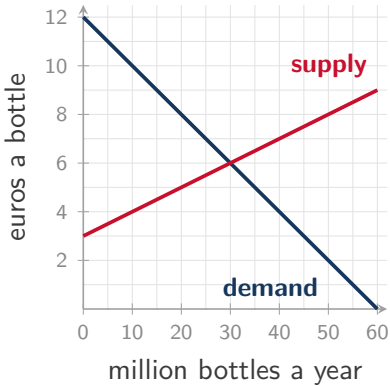
World price below home price: imports; buyers gain, makers lose less



Buyers gain 2,640, home makers lose 1,200: the country gains 1,440 million euros a year.

1 thousand cars × 1 thousand euros a car = 1 million euros.

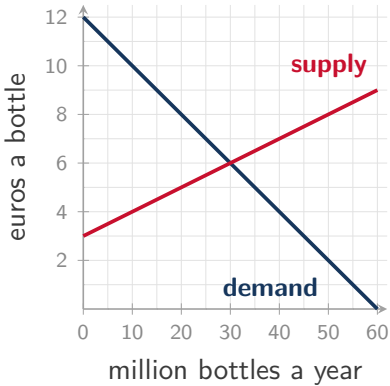
In pairs, three minutes: cava at home and abroad



Cava in a small country

1. No trade: price and bottles?
2. World price 8: export or import? How many?
3. At 8, who loses?
4. World price 4 instead?

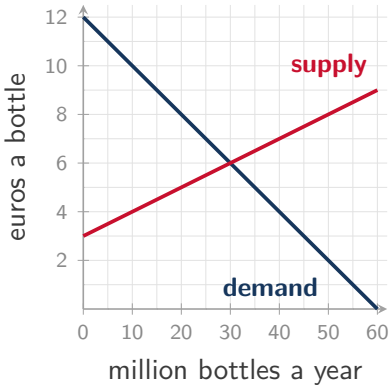
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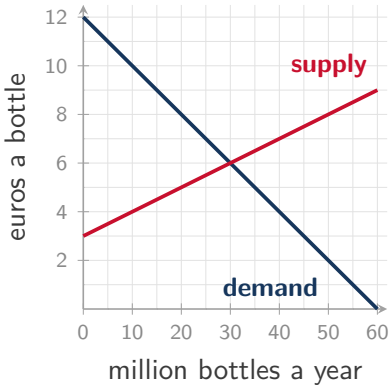
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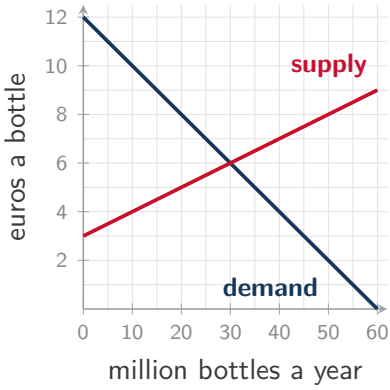
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   export 30 million: 50 made, 20 drunk
3. At 8, who loses?
   home buyers: pay 8, drink 20 million
4. World price 4 instead?

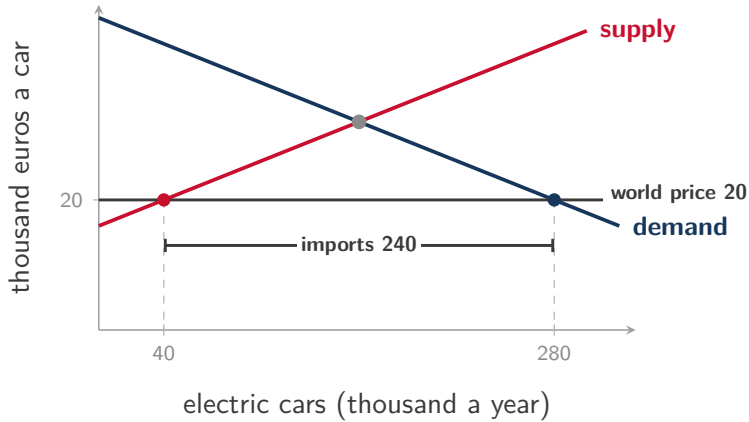
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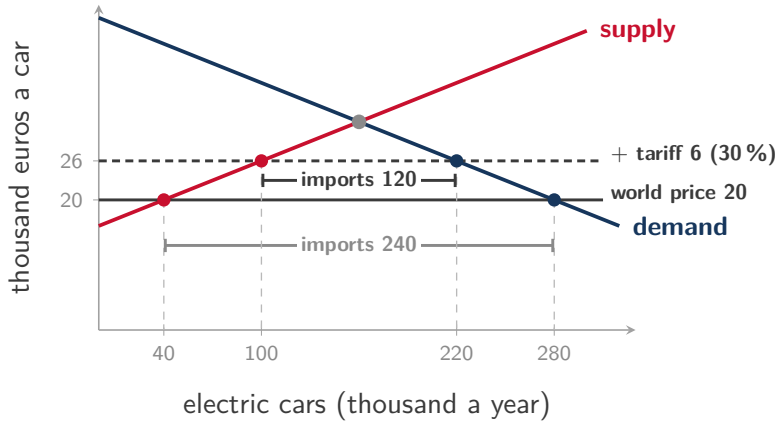
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   import 30 million: 40 drunk, 10 made

A tariff of 6 on imported cars: home price 26, imports halve

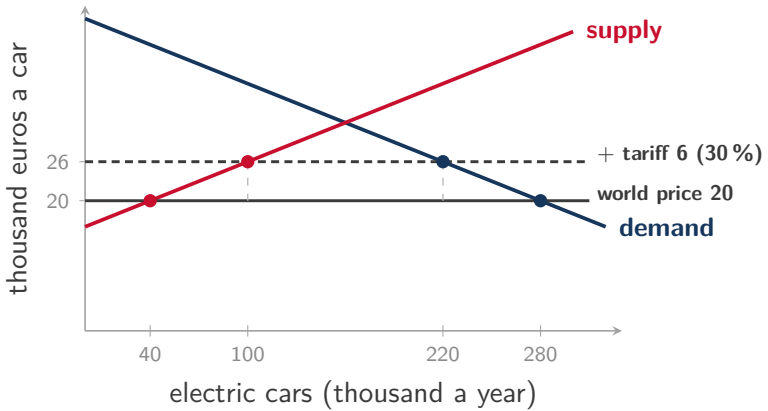


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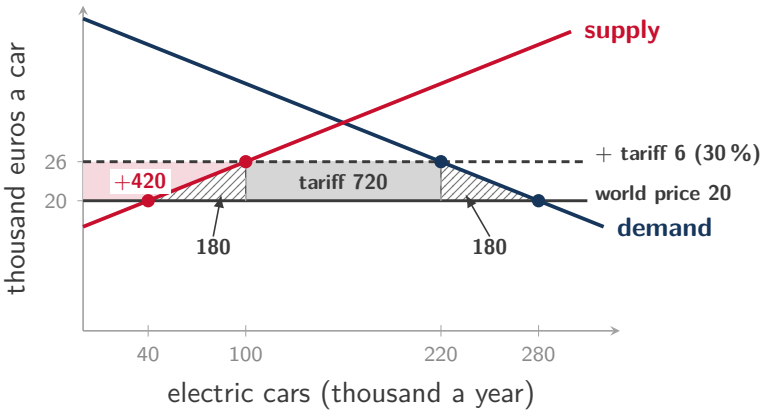


Home price 26: home makers 40 → 100, buyers 280 → 220, imports 240 → 120.

The tariff's bill: 360 million a year that nobody gets



The tariff's bill: 360 million a year that nobody gets



Buyers lose 1,500 = makers 420 + state 720 + 360 that nobody gets (million euros a year).

October 2024: the EU adds duties on electric cars made in China

EU's ordinary duty on cars (China included)	**10 %**
Plus, electric cars made in China:	
BYD	17.0 %
Geely	18.8 %
SAIC	35.3 %
other cooperating makers	20.7 %

February 2026: one China-made model exempted from the extra duty, for a minimum import price and a cap on numbers.

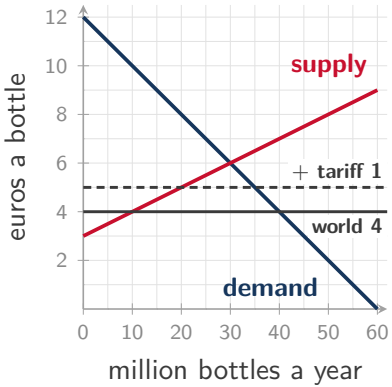
European Commission, IP/24/5589 (29 October 2024) and QANDA/24/3232 (12 June 2024); price undertaking accepted 10 February 2026.

Five arguments for limiting trade — and the catch

argument	the catch
jobs	export industries add jobs
national security	easily stretched to any industry
infant industry	hard to pick future winners
unfair competition	home buyers gain from cheap imports
bargaining chip	threats may have to be carried out

Each protects someone; home buyers pay, two triangles are lost.

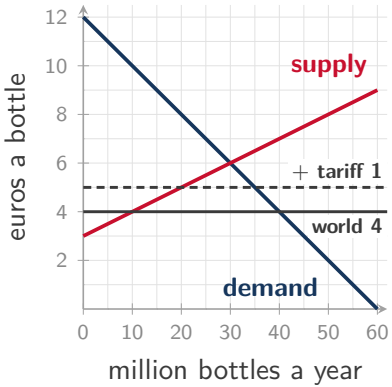
In pairs, three minutes: a tariff of 1 euro on imported cava



Cava: world price 4, tariff 1

1. Home price and imports?
2. Tariff revenue?
3. Surplus lost?
4. Who gains, and how much?

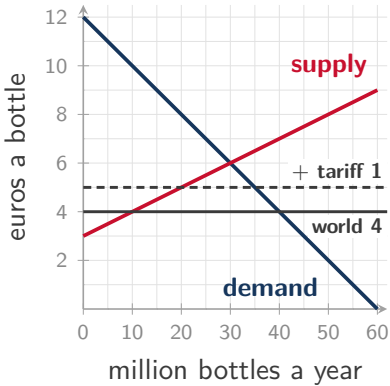
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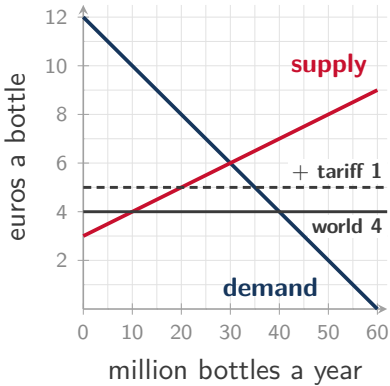
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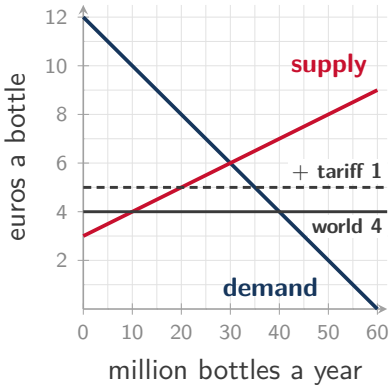
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   $\frac{1}{2} \times 1 \times 10 + \frac{1}{2} \times 1 \times 5 = 7.5$ million
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   revenue 15 million euros
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   ½ × 1 × 10 + ½ × 1 × 5 = 7.5 million
4. Who gains, and how much?
   home makers +15, the state +15

Practice at the board: a tax on driving lessons

Driving lessons in a town, a day; P in euros a lesson

Learners want $Q = 160 - 3P$ lessons · driving schools offer $Q = P$
The town charges the schools 4 euros a lesson

1. No tax: price and lessons?
2. With the tax: lessons, price learners pay, price schools keep?
3. Tax revenue and deadweight loss?
4. The 4 euros: how much more do learners pay, how much less do schools keep?

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   40 euros, 40 lessons
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   37 lessons; learners pay 41, schools keep 37
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   revenue $4 \times 37 = 148$; loss $\frac{1}{2} \times 4 \times 3 = 6$ (euros a day)
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4. The 4 euros: how much more do learners pay, how much less do schools keep?
   learners 1 more, schools 3 less:
   per euro, learners drop 3 lessons, schools only 1

Practice at the board: dearer pancakes (optional)

A pancake stand raises its price

Price 4 $\rightarrow$ 6 euros · pancakes sold a day 300 $\rightarrow$ 180

1. Price elasticity of demand, midpoint method (as a positive number)?

2. Revenue a day: up or down?

Practice at the board: dearer pancakes (optional)

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   $(120 \div 240) \div (2 \div 5) = 50\,\% \div 40\,\% = 1.25$: elastic
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   $(120 \div 240) \div (2 \div 5) = 50\,\% \div 40\,\% = 1.25$: elastic
2. Revenue a day: up or down?
   down: $4 \times 300 = 1{,}200 \rightarrow 6 \times 180 = 1{,}080$ euros

A tax, a cap or a tariff shrinks a market; twice the wedge, four times the loss.

A binding cap: cheaper for some, a shortage for others.

Open borders: the country gains, some lose; a tariff takes part of it back.

Monday: when a market hurts people outside it.

Before Monday 26 October

- Monday: market failures, externalities and public goods
- Read Mankiw & Taylor: Ch. 9, pp. 204–216; Ch. 8, pp. 189–197 (pp. 197–199 optional)
- Problem set 2 (Block 3): handed out Monday; due Monday 2 November
- Practise in Sextant: the units the course page lists



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